

ZACHARY TIANYI TANG

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RESEARCH INTERESTS

Sample-efficient robot learning, human-in-the-loop policy adaptation, and contact-rich/deformable-object manipulation.

EDUCATION

University of Toronto

Honours B.Sc., Computer Science; GPA: 3.77/4.00

Toronto, ON

Expected May 2028

RESEARCH & ENGINEERING EXPERIENCE

RLLAB, Yonsei University (Prof. Youngwoon Lee)

Research Intern

Seoul, South Korea

May 2026 - Aug 2026

- Leading **first-author research** on fine-tuning and evaluating pretrained **Vision-Language-Action (VLA)** policies with limited human feedback, studying when interventions most effectively correct robot behavior.
- Built and operated a **two-arm teleoperation setup** for real-world robot data collection, helping develop the **control framework** and collecting **intervention and demonstration data**.
- Implemented and evaluated **offline post-training (RL-Token-style)** using failure and intervention data, earning **top-8 finishes** in both tracks of the **RSS 2026 Robotics Foundation Models Challenge**.

Dynamic Optimization & RL Lab, University of Toronto (Prof. Chi-Guhn Lee)

Researcher

Toronto, ON

Sep 2025 - Present

- Co-authored **WireCraft (CoRL 2026 Spotlight; DOM-R3 Workshop Oral, IROS 2026)**, a **flexible-wire manipulation benchmark** showing that combining simulated and real demonstrations outperformed real-only training on physical robots.
- Deployed, fine-tuned, and evaluated $\pi_{0.5}$ **VLA policies** with **OpenPI**, plus **ACT and multi-task DiT policies**, for UR5 wire manipulation; built **LeRobot-compatible datasets** and scalable training and inference pipelines.
- Built and benchmarked **Isaac Lab environments** for articulated and deformable wires, designing clip-routing demonstrations, rewards, and success metrics while debugging rare **PhysX contact failures** across millions of RL simulation steps.

Human Computer Lab (Robotics Startup)

Software Engineer

Remote

Jan 2026 - Mar 2026

- Built **physical-digital interaction prototypes** for the LeLamp robot, connecting Raspberry Pi-based robot control to real-time browser games over **WebSockets** and mapping physical robot motion into interactive environments.
- Developed and deployed **end-to-end embodied interaction demos**, including an onboard camera system and real-time robot/game control loop for internal pilot testing.

Intelligent Robot Learning Lab, University of Alberta (Prof. Matthew Taylor)

Research Lead

Remote

Sep 2025 - Present

- Led an undergraduate research team studying how AI agents coordinate when each agent has incomplete information, building **JAX/JaxMARL training environments** and communication experiments.
- Built an **IsaacLab simulation** with two **Franka robot arms** to study how robots share limited information while coordinating tasks, integrating high-level action controllers, local observations, and learned messaging.
- Received the **People's Choice Award** at the **2026 Canadian Undergraduate Conference on AI (CUCAI)**.

PROJECTS & LEADERSHIP

Heimer Academy | [React](#), [Vite](#), [FastAPI](#), [Supabase](#), [Amazon SageMaker](#), [Amazon Bedrock \(Devpost\)](#) ([GitHub](#))

- 1st Place** — **AWS × Riot Games Rift Rewind (\$10,000 USD + \$10,000 AWS credits)**
- Built a **League of Legends champion recommendation system** using gameplay patterns across **tens of thousands of player profiles** and similarity analysis over **600+ champion abilities**.

University of Toronto Machine Intelligence Student Team (UTMIST)

Director of Academics

Sep 2026 - Present

Director of Research Collaborations

Jun 2025 - Aug 2026

- Led **15+ researchers** across two AI research teams and previously led a 10+ person engineering team delivering an **end-to-end AI system** for an external client, while establishing a research collaboration with the **University of Alberta**.

TECHNICAL SKILLS

Programming: Python, C, JavaScript, TypeScript

Robotics / ML: PyTorch, JAX, LeRobot, Isaac Sim, Isaac Lab, OpenCV

Systems / Tools: CUDA, TensorRT, SLURM, Docker, Git, Linux, Bash